Kennesaw State University 2026-2027 Undergraduate Catalog

Courses

Contract All Courses |

Software Engineering

- ### SWE 3313: Introduction to Software Engineering

3 Credit Hours

Prerequisite: (CSE 1322 and CSE 1322L)

Description

This course provides an overview of the software engineering discipline, introducing the student to the fundamental principles and processes of software engineering. This course highlights the need for an engineering approach (both personal and team) to software with understanding of the activities performed at each stage in the development cycle. In this course, students will perform requirements analysis, design, implementation and testing. The course presents software development processes at the various degrees of granularity. Students will become aware of libraries of standards (IEEE, ACM, SWEBOK, etc.).

- ### SWE 3623: Software Systems Requirements

3 Credit Hours

Prerequisite: (SWE 3313 and (CSE 2300 or MATH 2345)) or CPE 3000

Description

This course covers engineering activities related to the definition and representation of software system requirements. Topics include the elicitation, analysis, specification and validation of software system requirements. Emphasis is on the application of processes and techniques of requirements engineering. Projects focus on current analysis methods and supporting tools for specification, organization, change management, traceability, prototyping, and validating requirements.

- ### SWE 3633: Software Architecture and Design

3 Credit Hours

Prerequisite: SWE 3313 or CPE 3000

Description

This course covers the fundamental design principles and strategy for software architecture and design. Architectural styles, quality attributes, design notations and documents, reference architecture, domain specific architecture in architecture process and pattern-oriented design, component-oriented design, and interface design in detailed design process are discussed.

- ## SWE 3643: Software Testing & Quality Assurance

3 Credit Hours

Prerequisite: SWE 3313 or CPE 3000

Description

This course will show how software quality assurance and configuration management are performed and how software process improvement is maintained in order to assure the highest possible quality. Topics include software process metrics and their use in QA, testing approaches, methods and techniques. Development of QA plans, reviews, inspections and audits will be done.

- ## SWE 3683: Embedded Systems Analysis and Design

3 Credit Hours

Prerequisite: CS 3305

Description

The analysis and design course focuses on using modern methods, techniques, and tools for specification and design of embedded systems. Topics include analytical methods such as RMA, development methods such as HOOD, and notations like UML, Petri-nets, etc. are covered. Performance evaluation based on modeling and simulation techniques is also covered. This is a project based course.

- ## SWE 4324: User-Centered Design

3 Credit Hours

Prerequisite: SWE 3313

Description

This course presents the fundamental knowledge, processes, skills, and practices leading to the user-centered design (UCD) of (computer) systems and applications. UCD is not limited to the design of software systems but applies to the plan and development of any objects. This course will define the important concepts in UCD,

cognition and its relationship with UCD, and two views of human-centered design. Usability engineering techniques are covered leading to improved system effectiveness in supporting use of computers, user learning, diversity in interaction styles, and individual versus group work.

- ## SWE 4490: Special Topics

3 Credit Hours

Description
This course offers advanced topics in Software Engineering selected by the Department. The course covers special topics at the senior level that are not covered in the regular course offerings.

- ## SWE 4633: Cloud Software Development

3 Credit Hours

Prerequisite: CS 3305
Description
This course discusses the fundamental concepts and tools of cloud-based application development. Students will learn how to develop apps on top of the cloud with an understanding of various contemporary cloud providers (e.g., Amazon Web Services, Google Cloud, or Microsoft Azure). Students will do hands-on projects focused on building cloud-run applications using their platform of choice.

- ## SWE 4663: Software Project Management

3 Credit Hours

Prerequisite: SWE 3313 and STAT 2332
Description
This course focuses on organizational and technical roles in software engineering. Models of software engineering life cycle, software maturity framework, strategies of implementing software, software process assessment, project planning principles and tools, software configuration management, managing software quality and usability, leadership principles and legal issues will be covered. A required team project combines technical and managerial techniques of software design and development.

- ## SWE 4713: SWE Application Domain

3 Credit Hours

Prerequisite: SWE 3623
Concurrent: SWE 3643 and SWE 4663

Description

Students work as part of a team to develop solutions to problems posed by either internal or external customers in a specific SWE Application Domain. The purpose of the course is for students to gain an understanding of the selected application domain and its use of software to support functions/operations within that domain. Application domain selection is done every term from a variety of industrial domains including Security, Gaming, Automotive, Aerospace, Military, Finance and Commerce. Problems may require considerable software development or evolution and maintenance of existing software products. The course culminates with the completion and presentation of an increment of the project solution.

- ## SWE 4723: Undergraduate Research Methods

3 Credit Hours

Prerequisite: SWE 3313 or instructor permission.

Description

High Impact Practices (HIPs) are recognized as an important tool that can transform the educational experience of a student. Among the different possible HIPs, undergraduate research is an important activity that allows for training creativity and critical thinking. Being a good researcher depends on many factors and requires discipline and the knowledge of the research process and related methodologies. This course will provide a comprehensive introduction to give students the tools to create a research project, including research methodologies and foundational research theories and protocols. This course will also teach students how to write the results of their research and how to present them.

- ## SWE 4724: Software Engineering Capstone Project

4 Credit Hours

Prerequisite: SWE 3623
Concurrent: SWE 3643 and SWE 4663

Description

This is the capstone project course and constitutes a major design experience. The course focuses on a team project comprising the development of a realistic software system during all phases of the software development life cycle. Topics include software project management, design, verification and validation, development, evolution and quality assurance. Current methods, techniques, and software tools are utilized in the development of the project.

- ## SWE 4743: Object-Oriented Development

3 Credit Hours

Prerequisite: CS 3305

Description

This course involves engineering activities related to the analysis, design, and implementation of object-oriented software systems. Topics include modeling foundations, requirements specification and documentation, design concepts and strategies, and OOAD methodologies with an emphasis on UML. The course includes a major project utilizing current analysis and design methods and tools implemented in a contemporary IDE.

- ## SWE 4783: User Interaction Engineering

3 Credit Hours

Prerequisite: SWE 3313 or SWE 4324

Description

This course covers the basic theories and principles applied to the design of human-computer interfaces. It introduces students to the basic concepts used in designing interactions. It will cover the basic theory and methods that exist in the field. Case studies are used throughout the readings to exemplify the methods presented and to lend a context to the issues discussed. This course will focus on the semiotic engineering approach to human-computer interaction, in which the interaction is analyzed in terms of communication between the designers of the artifact and the user.

- ## SWE 4803: Independent Study

3 Credit Hours

Prerequisite: Program Coordinator and Faculty approval

Description

Independent study/project under the direction of a member of the graduate faculty. Course description will vary.

- ## SWE 4994: Agile Development and Prototyping

3 Credit Hours

Prerequisite: ENTR 3001

Description

This course introduces the essentials of software engineering processes for the engineering and building of a minimum viable product (MVP). The emphasis is on the applicability of Agile processes in the software life cycle and high-fidelity prototyping. Students gain an understanding of the commonly used Agile Scrum, as well as

its working mechanics. They will put into practice their knowledge by building MVPs using Agile in the course for scale.

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